Box-Cox gamma sweep | Val All42 variants: common ICs [0, 9, 10, 14, 17, 19, 24, 33, 41, 46, 63, 71, 81]

Repaired | Cantera vs NN NRMSE | Median of IC-wise scores; completion counts use all18 ICs

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
1 ANN10	0.05	0.125	0.352	0.998	0.86	0.421	3.63	0.187	0.964	1.54	0	18/18
1 ANN10	0.1	0.112	0.46	1.19	0.825	0.405	3.22	0.232	1.02	1.56	0	18/18
1 ANN10	0.2	0.0733	0.529	0.92	0.964	0.358	1	0.179	0.995	1	0	18/18
1 ANN10	0.3	0.0245	0.309	0.847	0.773	0.13	0.846	0.0548	1.09	1.01	0	18/18
1 ANN10	0.5	0.0686	0.767	0.997	0.915	0.17	0.954	0.111	1	1.06	0	18/18
1 ANN10	1	0.284	1.47	1.46	0.995	0.932	0.999	0.693	0.999	55.9	0	18/18
2 P-only DeepONet10	0.05	0.209	0.829	0.972	1.05	0.268	3.07	0.283	0.941	1	0	18/18
2 P-only DeepONet10	0.1	0.0559	0.209	0.91	0.953	0.082	0.952	0.112	0.966	0.977	0	18/18
2 P-only DeepONet10	0.2	0.0722	0.782	0.897	0.956	0.0728	0.611	0.117	0.97	0.986	0	18/18
2 P-only DeepONet10	0.3	0.0593	0.441	1.13	1.1	0.194	1.36	0.0669	1	7.02	0	18/18
2 P-only DeepONet10	0.5	0.0975	0.916	0.985	0.921	0.659	0.532	0.119	0.969	0.992	0	18/18
2 P-only DeepONet10	1	0.215	1.05	1.2	2.14	0.841	0.969	0.249	1.06	20.8	0	18/18
3 ANN10 + P-only DeepONet10	0.05	0.162	0.539	0.959	0.982	0.228	3.04	0.255	1.07	1.47	0	18/18
3 ANN10 + P-only DeepONet10	0.1	0.0396	0.31	0.9	0.971	0.0564	0.63	0.0531	0.613	1	0	18/18
3 ANN10 + P-only DeepONet10	0.2	0.0427	0.705	0.872	0.797	0.134	0.214	0.073	0.985	1.01	0	18/18
3 ANN10 + P-only DeepONet10	0.3	0.0386	0.392	0.761	0.758	0.125	0.787	0.0457	1	1	0	18/18
3 ANN10 + P-only DeepONet10	0.5	0.053	0.518	1.92	0.821	0.253	0.913	0.0702	0.97	1.22	0	18/18
3 ANN10 + P-only DeepONet10	1	0.154	1.41	1.42	1.37	0.761	0.476	0.46	2.61	41.2	0	18/18
4 T/P DeepONet10	0.05	0.222	0.656	1	1	0.67	1	0.247	1.26	1.5	0	18/18
4 T/P DeepONet10	0.1	0.0711	0.48	1	0.813	0.124	0.957	0.108	0.999	0.999	0	18/18
4 T/P DeepONet10	0.2	0.106	0.768	0.709	1	0.319	0.396	0.168	1	1	0	18/18
4 T/P DeepONet10	0.3	0.0827	0.298	0.899	0.971	0.161	0.974	0.214	0.999	1	0	18/18
4 T/P DeepONet10	0.5	0.52	0.7	3.17	8.27	0.687	2.29	0.309	26.7	28.9	0	13/18
4 T/P DeepONet10	1	0.236	2.57	1	1.82	0.988	0.549	0.36	0.998	30.8	0	18/18
5 ANN10 + T/P DeepONet10	0.05	0.295	0.41	2.29	0.986	0.422	5.41	0.303	1.96	2.06	0	18/18
5 ANN10 + T/P DeepONet10	0.1	0.0529	0.21	0.794	0.646	0.0804	0.924	0.0835	0.825	0.804	0	18/18
5 ANN10 + T/P DeepONet10	0.2	0.0438	0.313	0.448	0.834	0.125	0.19	0.0768	0.977	1	0	18/18
5 ANN10 + T/P DeepONet10	0.3	0.0158	0.253	0.367	0.255	0.075	0.135	0.0296	0.6	0.79	0	18/18
5 ANN10 + T/P DeepONet10	0.5	0.0654	0.942	1.04	1	0.422	0.997	0.208	1.12	1.6	0	18/18
5 ANN10 + T/P DeepONet10	1	0.0936	0.874	2.75	0.794	0.744	0.531	0.211	0.818	23.4	0	18/18
6 T/P/ϕ DeepONet10	0.05	0.421	0.708	5.08	1.31	0.376	4.6	0.412	4.16	2.62	0	18/18
6 T/P/ϕ DeepONet10	0.1	0.155	0.494	1	1	0.252	1	0.135	0.998	1	0	18/18
6 T/P/ϕ DeepONet10	0.2	0.0426	0.331	1	0.644	0.131	0.602	0.0734	0.839	0.945	0	18/18
6 T/P/ϕ DeepONet10	0.3	0.576	1.2	1	1.19	0.274	1.09	0.325	1	1.8	0	18/18
6 T/P/ϕ DeepONet10	0.5	0.206	1.47	1	1	0.83	1	0.495	1	1	0	17/18
6 T/P/ϕ DeepONet10	1	0.142	1.13	3.27	1.79	0.62	0.619	0.194	1.01	9.84	0	18/18
7 ANN10 + T/P/ϕ DeepONet10	0.05	0.193	0.443	0.979	0.988	0.4	1.01	0.437	1.55	2.49	0	18/18
7 ANN10 + T/P/ϕ DeepONet10	0.1	0.0606	0.246	0.821	0.424	0.125	0.947	0.113	0.844	0.915	0	18/18
7 ANN10 + T/P/ϕ DeepONet10	0.2	0.0167	0.168	0.445	0.218	0.0785	0.307	0.0511	0.824	1	0	18/18
7 ANN10 + T/P/ϕ DeepONet10	0.3	0.065	0.772	0.91	1	0.103	0.997	0.184	0.999	1	0	18/18
7 ANN10 + T/P/ϕ DeepONet10	0.5	0.0595	1.91	0.952	0.811	0.775	0.765	0.247	0.982	0.99	0	18/18
7 ANN10 + T/P/ϕ DeepONet10	1	0.255	1.9	3.12	1.28	0.729	0.527	0.382	0.996	12.3	0	18/18

NRMSE = RMSE(Cantera,NN) / RMS(Cantera). Exact user util.py default; equal time-point weights; raw ratio without ×100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Nonzero constant N2 is valid.
Gamma is the Box-Cox exponent; soft constraint is absent. New models stop at10000 updates or150 seconds; actual endpoints are exported. Cantera + six gamma curves.